Predicting ΔΔG of protein–protein binding from sequence: how much of a fine-tuned ESM-2's accuracy is leakage?

Johannes Venezian September 2026 Code and full logs: github.com/DermIQ-admin/esm2-ddg-binding

Fine-tuning ESM-2 to predict the binding cost of a single interface mutation reaches **Spearman ρ = 0.66** when the test set is drawn at the mutation level, and **ρ = 0.18** when no protein complex is allowed to appear in both training and test. On that honest split a **57-feature substitution-chemistry model with no language model in it** is statistically indistinguishable from the fine-tuned network — and its own leaky-to-honest drop is **four times smaller**, because its features cannot identify which complex they are looking at. Adding one structural annotation beats every ESM-2 variant tried here by a wide margin. The gap between those numbers, and what closes it, is what this report is about.

1. Problem — and why it is a reasonable proxy for antibody optimization

ΔΔG of binding is the quantity an affinity-maturation campaign searches over: $\Delta\Delta G = \Delta G_{mut} - \Delta G_{WT}$, with $\Delta G = RT \cdot \ln K_d$, so positive means the mutation weakens the interface. Ranking candidate interface substitutions by predicted ΔΔG is the primitive a binder-engineering loop needs before committing to expression and SPR: it need not be calibrated in kcal/mol, it has to put the right mutations at the top of the list — which is why **Spearman is the primary metric here, with RMSE reported alongside rather than instead**. The sequence-only formulation is the deliberately harder and more useful one, since it applies to the many antibody–antigen pairs with no solved co-crystal, and SKEMPI 2.0 makes it measurable: experimental K_d for wild-type and mutant complexes of known structure.

My doctoral work approached the same interface from the opposite side. In *Nature Communications* (2024) we showed that co-translational assembly pathways in *S. cerevisiae* are governed by how interface energy is distributed across a complex rather than by its total magnitude; the follow-up in *Cell* (2025), with the Levy lab, found that structural signatures — subunits mutually stabilising one another — predict which pairs assemble. Both are structure-first results: the prediction came from a solved or modelled complex, and my own contribution was the experimental half that tested it, by selective ribosome profiling and RIP-qPCR on strains built for the purpose.

That is why the sequence-only formulation here is the complementary question rather than a restatement of it. The published work presumes a structure and asks what its energetics predict; this asks how much of the same information survives once the structure is removed — the situation for most antibody–antigen pairs. The answer §4 arrives at is that less survives than reported numbers in this area suggest, and the fact that one coarse structural annotation recovers more than any amount of sequence pretraining is the structure-first conclusion arriving again from the opposite direction.

2. Data, and the split that decides everything

Dataset. SKEMPI 2.0 ships 7,085 mutation records; restricting to single-point mutations leaves 5,112. A further **127 rows are dropped because their affinity is a bound, not a measurement** — SKEMPI's convenience columns `Affinity_*_parsed` strip the leading < or > and return a bare number, so filtering on missing values alone silently keeps "no detectable binding" entries as if they were real K_d. The filter runs against the raw affinity columns, where the marker survives. Dropping 156 further rows with unusable K_d gives **4,829 mutations across 316 complexes**. Sequences are reconstructed per complex from the residue-mapping files SKEMPI ships with each structure; the wild-type residue named by each mutation code matches the reconstruction for **5,112 / 5,112 rows (100%)**, asserted at parse time because an off-by-one here is what this pipeline is most exposed to. A ≤2,048-token length policy then excludes two large multi-chain assemblies, leaving **4,814 mutations across 314 complexes**. Labels skew hard toward destabilization (3,612 positive vs 1,050 negative; mean +0.96, median +0.56, sd 1.71 kcal/mol) with **11.0% of rows beyond |ΔΔG| > 3** — hence Huber loss rather than MSE.

The split. This is what determines whether any number below means anything. Mutations of the same complex are highly correlated — SKEMPI contains dozens of alanine scans of the same interface. Under a random row-level partition, nearly every complex in the test set also appears in training, so the model can memorise a complex from its training mutations and then "predict" held-out mutations of that same interface. Memorising and learning ΔΔG become indistinguishable, and both register as progress. **This project therefore partitions groups, never rows**, and reports three definitions side by side rather than choosing one:

Split definition	Groups	What it holds out
`split_mutation`	—	Naive row-level partition. **Leaks by construction.** Reported as the contrast case, never the headline.
`split_pdb_id`	313	Whole structures. The literal reading of "no complex in two splits".
`split_hold_out_proteins`	152	SKEMPI's curated protein-pair grouping. Strictest — also catches one protein pair appearing under several PDB entries.

Assignment is size-aware rather than a uniform sample of groups: SKEMPI's group sizes span three orders of magnitude, and sampling uniformly produced a 4.7% validation set where 15% was intended — too thin to select a model on. Groups are instead shuffled under a fixed seed, taken largest-first, and each given to whichever split is furthest below its row target, landing all three definitions at 70.0 / 15.0 / 15.0 by row count with groups kept atomic. The invariant is therefore structural rather than merely tested, and it is also tested: 14 `pytest` cases assert that no group crosses a split boundary. **The cost of size-aware assignment, stated rather than buried:** filling `train` first sends the large, well-studied complexes there (121 training complexes vs ~96 each in val and test), so val and test hold smaller, less-studied ones — arguably the realistic generalization target, but a structured train/test difference rather than a purely random one.

3. Model and setup

A **siamese ESM-2**: one shared backbone encodes the wild-type and the mutant complex. Weight sharing is the point — it puts both embeddings in one space, so their difference is meaningful. Each is mask-aware mean-pooled, and the head sees `concat[wt, mut, mut - wt]`, since ΔΔG is a question about what changed. Both partner chains are concatenated into one sequence, so the backbone sees both sides of the interface at once.

BACKBONE
`facebook/esm2_t30_150M_UR50D` — 30 layers, hidden 640

HEAD
Linear(1920→256) → ReLU → Dropout(0.1) → Linear(256→1)

TRAINABLE PARAMETERS
148.6M, full fine-tune — of which the head is 0.49M

LOSS / OPTIMIZER
Huber δ = 1.0; AdamW, lr $2{\times}10^{-5}$, no schedule

PRECISION
bf16 autocast, no GradScaler (bf16 keeps fp32's exponent range)

BATCHING
By **token budget** — 4,096 padded tokens, not a fixed example count

EPOCHS / SELECTION
≤10, early stopping patience 3; best **validation Spearman**, not val loss

HARDWARE
1× RTX 4090 24 GB, Windows 11, PyTorch 2.13 / CUDA 13.0

WALL CLOCK
181 s/epoch, ≈26 min per run; 8.5 GPU-hours for all 20 replicate runs

REPLICATES
5 per method per split — 3 training seeds at fixed partition, 3 split seeds at fixed training seed

150M rather than 650M — measured, not assumed. 150M peaks at 10.9 GB; 650M needs 26.9 GB on a 24 GB card even with gradient checkpointing, because the siamese design calls the backbone twice per step, and runs **33× slower** once traffic spills to host memory — silently, since CUDA on Windows falls back to shared system memory rather than raising `OutOfMemoryError`. Two baselines keep the fine-tuned model from failing silently: zero-shot masked-marginal fixes the floor, a frozen backbone with a trained head measures what the representation already knows. And because the three split definitions disagree about which rows are held out, **a separate model is trained per definition**, the harness marking the other two contaminated.

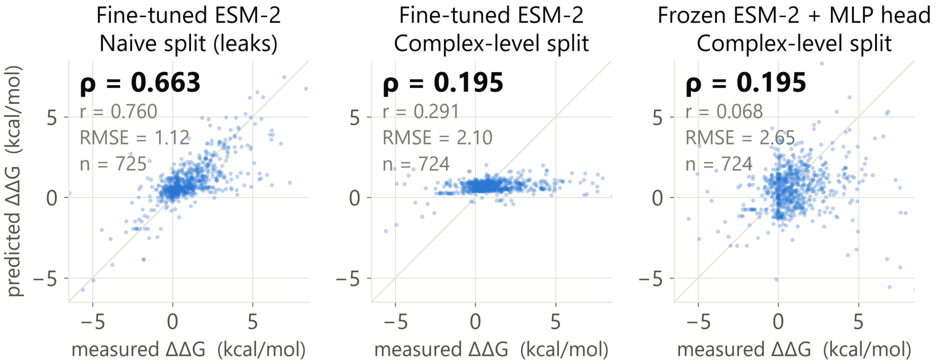

Figure 1 — the same model, the same data, two split definitions. Predicted vs measured ΔΔG on held-out test rows; the diagonal is $y = x$. **Left:** the naive split, where the same complexes appear in training. **Centre:** identical model and hyperparameters under a complex-level split — the prediction range collapses toward the training mean, which is what ρ = 0.20 looks like. **Right:** a frozen backbone with a 0.49M head on the same partition, matching it. Panels show the **median** of five replicates, not the best; the probe is drawn on the same partition as the fine-tuned model beside it.

4. Results — test Spearman ρ, mean ± sd over 5 replicates

Model	Trainable params	`split_mutation` naive — leaks	`split_pdb_id` honest	`split_hold_out_proteins` honest, strictest
Zero-shot ESM-2 150M (masked-marginal)	0	−0.082	+0.005	−0.097
Frozen ESM-2 + ridge	1.9K	0.473 ± 0.021	0.053 ± 0.057	0.122 ± 0.102
Frozen ESM-2 + MLP head	0.49M	0.623 ± 0.024	**0.223 ± 0.020**	**0.272 ± 0.053**
Fine-tuned ESM-2 150M	148.6M	**0.660 ± 0.021**	0.178 ± 0.059	0.167 ± 0.091
LoRA r=8 on the same backbone	1.11M	not run	0.211 ± 0.080	not run
— no language model below this line —				
Substitution chemistry + ridge	57 feat.	0.330 ± 0.029	0.192 ± 0.058	0.244 ± 0.009
Substitution chemistry + GBM	57 feat.	0.342 ± 0.028	0.173 ± 0.050	**0.272 ± 0.031**
…+ interface location [†]	62 feat.	0.483 ± 0.018	**0.389 ± 0.012**	**0.478 ± 0.017**

[†] **Not sequence-only.** `interface_location` is SKEMPI's structural annotation of the mutated residue (core / support / rim / interior / surface). Every other row uses sequence alone; this one is kept in the same table because it answers a question the others cannot, but it is not a like-for-like comparison and is never quoted as one.

Zero-shot is a single deterministic, ranking-only run, so it has no error bars — but its flatness is the control that matters: **the three test sets are of similar intrinsic difficulty**, so the gaps across the columns come from training, not from one split being easier.

That floor is explained, not merely reported: scoring the mutated chain *alone* and inside the full complex agree at ρ = 0.888 — the masked-marginal is blind to the binding partner, scoring whether a residue belongs in its chain rather than whether it holds an interface together.

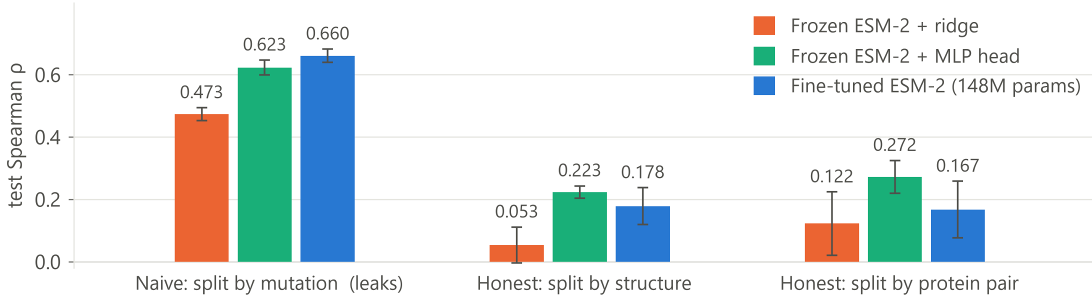

Figure 2 — the leakage gradient. Test Spearman by method under each split definition; bars are the mean of 5 replicates, whiskers ±1 sd. Note the *within-group* order too: on the leaky split more trainable parameters help monotonically; on both honest splits that ordering disappears. LoRA is absent from this figure by design — it was run on `split_pdb_id` only, so it has no bars to place in the other two groups.

Fine-tuning wins on exactly one split — the leaky one

Replicates are paired on identical partitions, so this compares fine-tuning against the frozen MLP probe on the same data rather than as two independent samples:

Split definition	Mean Δρ (fine-tune − probe)	95% CI	*p*	Verdict
`split_mutation` (leaks)	**+0.038**	[+0.003, +0.072]	**0.038**	Fine-tuning genuinely better
`split_pdb_id`	−0.045	[−0.133, +0.043]	0.229	Not distinguishable
`split_hold_out_proteins`	−0.105	[−0.256, +0.045]	0.124	Not distinguishable

The only split on which 148M trainable parameters demonstrably beat a frozen backbone plus a 0.49M head is the one that rewards memorisation. The training curves say why, across all five replicates rather than in one illustrative run. Under the naive split, validation holds other mutations of complexes already in training, so the best epoch is the **last** one in all five runs (val ρ 0.17–0.35 at epoch 0 rising to 0.62–0.68 at epoch 9) — it never signals a stop. Under `split_pdb_id` the best epoch is erratic (3, 3, 7, 9, 9); under the strictest split **three of five replicates peak at epoch 0**, before the model has learned anything transferable at all. The diagnosis is ordinary overfitting, not a bug: 148M parameters against 3,374 training rows from 121 complexes, ending at train ρ = 0.87 against test ρ = 0.22.

A second finding worth as much as the first: fine-tuning is 2–4× more variable than the probe (sd 0.048–0.113 vs 0.016–0.063), and fixing the random seed does not remove it — the identical configuration produced 0.148 and 0.225 on two different days, from CUDA kernel nondeterminism alone. A single fine-tuning run of this model is not a reportable number, which is why everything above is a 5-replicate mean. **LoRA was run against exactly this.** Training 1.11M parameters rather than 148.6M leaves the honest-split score unchanged (0.211 ± 0.080) but cuts *training-seed* variance ~5×, sd 0.074 → 0.014, the frozen probe's level. It does nothing for variance across *which complexes* are held out (0.048 → 0.093) — the one that limits a generalisation claim.

A baseline with no language model in it

Every method above is ESM-2, which leaves the obvious question open: what does a model with no learned representation get? The baseline uses 57 features a biochemist would write down in five minutes — BLOSUM62 score, changes in hydropathy, volume and charge, flags for proline/glycine/alanine, and one-hot residue identities. No structure, no evolution, nothing learned. **On the honest splits it is not distinguishable from ESM-2**, paired over the same five partitions: against the frozen probe, *p* = 0.148 and 0.987; against the fine-tuned network, *p* = 0.898 on `split_pdb_id` and **+0.106, *p* = 0.049 in the chemistry model's favour** on the strictest split.

It also shows why the naive split flatters ESM-2, rather than asserting it. A model built on substitution chemistry cannot tell which complex it is looking at — the features carry no complex identity — so it has nothing to memorise. How much each method loses when complexes stop being shared:

representation	leaky	honest	drop
ESM-2 fine-tune (148.6M)	0.660	0.178	**0.482**
ESM-2 MLP probe (0.49M)	0.623	0.223	**0.399**
ESM-2 ridge	0.473	0.053	**0.419**
Substitution chemistry + GBM	0.342	0.173	0.169
Substitution chemistry + ridge	0.330	0.192	0.138

Every representation that *can* encode complex identity loses 0.40–0.48. Every representation that cannot loses 0.14–0.17. **The size of the leakage effect tracks the capacity to memorise** — measured here rather than inferred, which is the strongest evidence in this report for the claim the whole thing is built around.

One structural annotation beats all of it. Adding `interface_location` takes the same model to **0.389** and **0.478** on the two honest splits, past every ESM-2 variant by +0.165 to +0.312, all $p \le 0.002$. That 5-level categorical scores ρ = 0.370 alone. The effect is what biophysics predicts, which is the main reason to believe it: mean ΔΔG runs +1.47 (interface core) → +1.21 (support) → +0.54 (rim) → +0.36 (protein interior) → +0.14 (surface). Two caveats: it is **not a sequence-only result** and is never compared as one —

what it establishes is the size of the prize for the structural features listed under limitations, which is larger than anything the language model contributed; and location is partly confounded with complex identity, since 137 of 316 complexes carry only one distinct location value.

The antibody–antigen subset

SKEMPI labels antibody–antigen complexes, so this breakdown is free — and it is the subset that matters most for binder engineering. It was pre-specified before any results existed, making it a planned comparison rather than a subgroup found by searching.

AB/AG test rows only (189–230 rows)	`split_pdb_id` — honest	`split_mutation` — leaks
Frozen ESM-2 + MLP head	**+0.264 ± 0.052**	+0.551 ± 0.042
Fine-tuned ESM-2 150M	**−0.065 ± 0.142**	+0.497 ± 0.083
Paired difference (fine-tune − probe)	**−0.330**, [−0.540, −0.119], p = 0.012	−0.054, [−0.121, +0.012], p = 0.086

Across all test rows, fine-tuning and the frozen probe are indistinguishable. **On antibody complexes they are not: fine-tuning is reliably worse**, by −0.330 (95% CI [−0.540, −0.119], p = 0.012), with the paired difference negative in all five replicates. Note what is *not* claimed: its own AB/AG correlation averages −0.065, but two of five replicates are positive and that mean is not distinguishable from zero (p = 0.362) — the replicates support the comparison, not the sign. Two further caveats: the strictest split holds only 6 AB/AG test rows against a 10-row minimum, so the metric is omitted there and the claim rests on `split_pdb_id` alone; and it is one subgroup among several — a pre-registered signal, not a confirmed mechanism.

5. Limitations

- **The honest numbers are modest (ρ ≈ 0.17–0.27), and they are the reported result.** A correctly attributed number on an honest split is worth more than an inflated one; the 0.66 is included to show what a split choice is worth.
- **Sequence-only** — and the cost is now measured, not assumed: one structural annotation more than doubles the chemistry baseline's honest-split score, past every ESM-2 variant here. Also **single-point only**, dropping 1,973 multi-point rows, so nothing here speaks to epistasis. No comparison against FoldX-style physics on its own terms is attempted.
- **Mean-pooling discards position** — a whole-sequence mean is a weak signal for a change of one residue in up to 2,048, plausibly the largest architectural limitation here. Separately, **127 inequality rows are dropped rather than used** (censored regression would use them properly), and val/test hold smaller, less-studied complexes as a side effect of size-aware assignment (§2).
- **Rows between 1,024 and 2,048 tokens (1.6%) exceed ESM-2's pretraining crop** and rotary embeddings extrapolate there without guarantee. Truncation was rejected because at a 1,024 cap the mutation site falls outside the window for 77 rows, making `mut - wt` exactly zero while the model still trains against a real label — silent rather than loud.

6. Next steps — reordered by what the chemistry baseline showed

- **Real structural features** from the PDBs rather than SKEMPI's annotation — relative solvent accessibility, interface contact counts, burial on complexation. The coarse 5-level proxy already beats every language model here, so the continuous versions are the obvious next measurement, and they would resolve the location/complex confound.
- **Combine chemistry with the embeddings.** Indistinguishable on the honest splits does not mean they carry the same information; one cheap fit tests whether the errors are correlated.
- **Attack split-seed variance, which LoRA did not.** LoRA fixed the training-seed kind and left this one untouched, and it is the kind that limits a generalisation claim. Layer-selective fine-tuning is the concrete candidate — AbTune reports 50–75% of layers beating full depth. LoRA also now unblocks 650M.
- **Position-specific pooling at the mutated residue,** concatenated with the mean-pooled context; the dataset already carries the token index. Lower priority than it was — the chemistry baseline suggests the ceiling may not be a pooling problem.
- **Diagnose the antibody-subset inversion** before trusting any antibody number: is CDR sequence variability being read as noise, or are the loops simply out of distribution for a general protein language model?

7. Related work

The closest sequence-only comparison is **MINT** (*Nature Communications*, 2026), which extends ESM-2 to encode interacting chains jointly through cross-attention and pretrains on STRING, reporting a 29% improvement on SKEMPI and the strongest published performance for a model using sequence alone. Its downstream setup — separate embeddings of the wild-type and mutant groups, aggregated and passed to an MLP — is close to the frozen probe that wins on the honest splits here, which suggests the architectural change rather than the fine-tuning is where its gain lives. **ProBASS** (Gurusinghe et al., 2025) is not sequence-only: it combines ESM-2 embeddings with structure-conditioned vectors from ESM-IF1. It needs a structure only of the wild-type complex, not one per variant — the authors note backbone positions are near-identical between wild type and mutant and reuse the wild-type embedding — so both it and `interface_location` in §4 rest on one structure. The difference is what each extracts from it: a learned per-residue embedding against a five-level categorical, and the five-level version already recovers much of the gain. **Seq2Bind** (*NAR Genomics and Bioinformatics*, 2025) fine-tunes ProtBERT, ProtT5, ESM-2 and a BiLSTM on SKEMPI 2.0, but evaluates interface-residue recovery under an N-factor metric rather than held-out ΔΔG regression, so it does not bear directly on the numbers here — though its own observation that the LSTM's predictions span only −0.368 to 0.07 kJ/mol across 6,063 complexes, about a tenth of a kcal/mol end to end, is the same range collapse visible in the centre panel of Figure 1. **AbTune** (Xu & Bonvin, *Briefings in Bioinformatics*, 2026) is the natural antibody comparison and takes the more interesting position: it fine-tunes at test time, per sequence, explicitly to partially bypass the need for generalization, and finds depth-selective tuning of 50–75% of layers beats full depth. That is a direct response to the failure mode measured in §4 — and the reason layer-selective fine-tuning is the first architectural item under next steps rather than the last.

On the question §2 turns on, the field has moved recently and in the same direction. **CATH-ddG** (*Bioinformatics*, 2025) states plainly that deep learning models may overestimate performance through data leakage, proposes train/test splits by CATH homologous superfamily, and notes that deep models beat physics-based approaches on easy mutations while generalizing poorly on hard ones. **SE3Bind** (bioRxiv preprint, 2026) tabulates the gradient directly: TopNetTree falls from $\rho = 0.85$ under a random K-fold split to 0.38 under an ECOD split, and Graphinity from 0.87 to 0.25 under CDR-loop clustering. The 0.660 → 0.178 measured here is the same phenomenon at the same magnitude, which is the main reason to read the honest column as the correct one rather than as a failed run. Two caveats on all of the above: none of these numbers are directly comparable to the results table in §4 — they differ in metric, in mutation scope, and in whether structure is used — and MINT's SKEMPI protocol in particular should be checked at the split level before its headline is set against anything here.

 Weights: `facebook/esm2_t30_150M_UR50D`. Every figure and table here regenerates from `results/metrics.json` via `report/make_figures.py` and `python -m src.replicates`.